

Lecture 4: Conditional Probability

BIOS 600 - Fall 2026

Welcome!

Take 1 piece of paper per 2 people.

Announcements

- HW 1 Due tonight at 11:59pm
- Lab 3 session Today at 3:30pm, due Friday at 11:59pm
 - Get started early!

Ensure your code is visible in your PDF document!

- For all assignments
- Go from this:

```
1 ggplot(data = cdc, aes(x = Region, y = InfantMortalityRate)) +  
2   geom_boxplot() + labs(title = "Infant Mortality Rate (%) by Region", xlab = "Regi
```

to this:

```
1 ggplot(data = cdc, aes(x = Region,  
2                       y = InfantMortalityRate)) +  
3   geom_boxplot() +  
4   labs(title = "Infant Mortality Rate (%) by Region",  
5         xlab = "Region",  
6         ylab = "Infant Mortality Rate (%)")
```

Reading

- Pagano and Gavreau: Sections 6.2
- [OpenIntro Statistics](#): Section 3.2

Overview

- Example & visual illustration of probability
- DeMorgan's Laws
- Conditional probability
- Independence and disjoint events

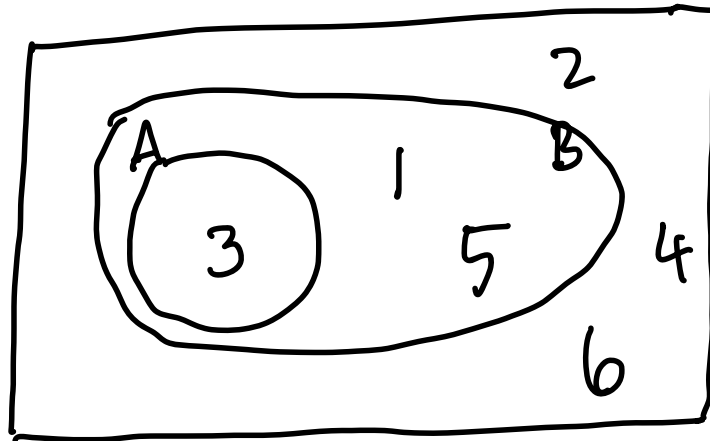
Example

Consider the experiment of rolling one fair six-sided die.

Let's define the following events:

- A : roll a 3
- B : roll an odd number

Let's draw out our experiment visually.



Practice

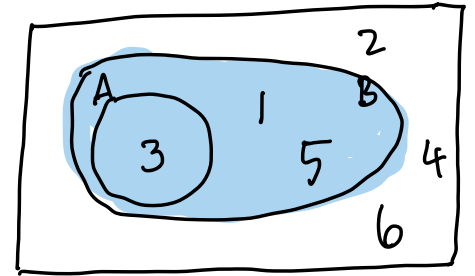
Using your illustration, determine the following:

a. What is the sample space of the experiment? $S = \{1, 2, 3, 4, 5, 6\}$

b. What is $P(A)$? $\frac{1}{6}$

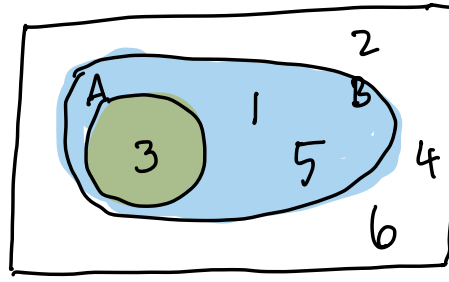
c. What is $P(B)$? $\frac{3}{6} = \frac{1}{2}$

d. What is the probability of A happening, given we know we've rolled an odd number?



$$P = \frac{1}{3}$$

Practice



$$S = \{1, 2, 3, 4, 5, 6\}$$

e. What is $P(A^C)$? $\frac{5}{6}$

f. List out the events in $A \cup B$. What is $P(A \cup B)$?

g. List out the events in $A \cap B$. What is $P(A \cap B)$? $\Rightarrow \frac{1}{6}$

h. Are the events A and B disjoint?

$$A \cap B = \{3\}$$

$$A \cup B = \{1, 3, 5\} \quad P(A \cup B) = \frac{3}{6}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{6} + \frac{3}{6} - \frac{1}{6}$$

$$= \frac{3}{6} = \frac{1}{2}$$

h) A and B are not disjoint
since $A \cap B \neq \emptyset$

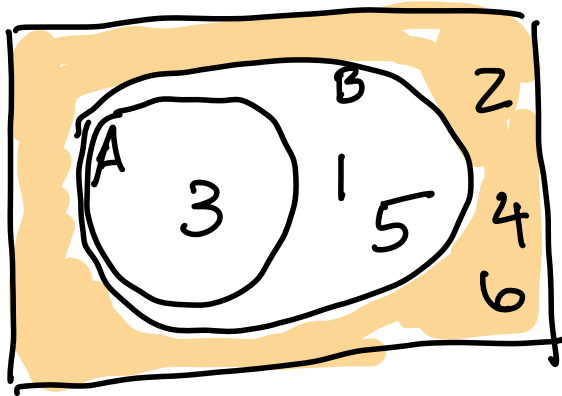
DeMorgan's laws



- Complement of union: $(A \cup B)^C = A^C \cap B^C$
- Complement of intersection: $(A \cap B)^C = A^C \cup B^C$

Practice

Verify the **Complement of union** law with our dice example.
(Draw it out!)



$$(A \cup B)^c = \{2, 4, 6\}$$

$$A^c = \{1, 2, 4, 5, 6\}$$

$$B^c = \{2, 4, 6\}$$

$$A^c \cap B^c = \{2, 4, 6\}$$

Conditional probability

Conditional probability

- **Conditional probability** is the probability an event will occur when another event has already occurred.

Examples

Examples come up all the time in the real world:

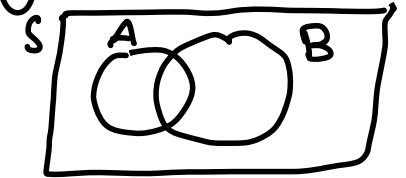
- Given that a mammogram comes back positive, what is the probability that a woman has breast cancer?
- Given that a 68-year old man has suffered four previous heart attacks, what is the probability he dies in the next five years?
- Given that a patient has a mutation in the CFTR gene, what is the probability their offspring will have cystic fibrosis?

Conditional probability

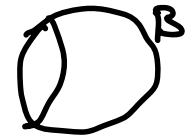
Definition: The conditional probability of event A given event B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

original:



new:



- How can we represent this visually? (Drawing)
- Takeaway: If we know B has already happened, our sample space effectively shrinks to only events that are contained within B .

The Multiplicative rule

We can rewrite the definition of conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \implies \underbrace{P(A \cap B) = P(A|B) \times P(B)}_{\text{Multiplicative Rule}}$$

Question

What does the multiplicative rule mean in plain English?

- The probability that A and B **both happen** is the probability that B happened, multiplied by the probability of A happened (given B already happened).

Independence

Events A and B are said to be **independent** when

$$P(A \cap B) = P(A) \times P(B)$$

or equivalently, $P(A|B) = P(A)$ or $P(B|A) = P(B)$

- In other words, if A and B are independent, then the probability of both happening is simply the probability of A happening times the probability of B happening. (Since B happening has nothing to do with A later happening, or vice-versa).

Example

- Experiment: Toss a fair 6-sided dice two times. We assume the two tosses are **independent** from each other.
- Let A = getting a 1 on the first toss. Let B = getting a 1 on the second toss.

- What is $P(A)$? $6/36 = 1/6$

- What is $P(B)$? $6/36 = 1/6$

- What is the $P(A \cap B)$? $= 1/36 = P(A) \cdot P(B)$

- What is $P(A|B)$? $1/6$

- What is $P(B|A)$? $= P(B)$
 $= 1/6$

$$S = \{ (1,1), (1,2), \dots, (1,6), \\ (2,1), (2,2), \dots, (2,6), \\ \vdots \\ (6,1), \dots, (6,6) \}$$

since
 A, B independent.

$$P(A|B) = P(A) \text{ since } A \perp B$$

Recall: disjoint events

- Definition: Two outcomes are **disjoint** or **mutually exclusive** if they cannot both happen.
- For example: If we roll a die, the outcomes 1 and 2 are disjoint since they cannot both occur.
- On the other hand, the outcomes 1 and “rolling an odd number” are not disjoint since both occur if the outcome of the roll is a 1.

Conditional probabilities and independence

A: died

B: non-coffee

Coffee drinking	Died? Yes	Died? No	Total
None	1039	5438	6477
Med-Low	4440	29712	29809
High	3601	24934	28535
Total	9080	60084	64821

What was the probability a randomly selected person in the study...

- ...died? $9080 / 64821 \approx 0.14$ $P(A)$
- ...died, given they were a non-coffee drinker? $1039 / 6477 \approx 0.16$ $P(A|B)$

Question

In this study, were dying and coffee drinking independent events?

Order matters!

What is the probability that a random person...

Coffee drinking	Died? Yes	Died? No	Total
None	1039	5438	6477
Med-Low	4440	29712	29809
High	3601	24934	28535
Total	9080	60084	64821

- ...was a high coffee drinker, given that they died? $\frac{3601}{9080} = 0.4$
- ...died, given that they were a high coffee drinker? $.13 = \frac{3601}{28535}$

Are these two probabilities the same? NO

Example: Smallpox

The `smallpox` data set provides a sample of 6,224 individuals from the year 1721 who were exposed to smallpox in Boston.

Doctors at the time believed that inoculation, which involves exposing a person to the disease in a controlled form, could reduce the likelihood of death.

Example: Smallpox

Each case represents one person with two variables: `inoculated` and `result`. The variable `inoculated` takes two levels: `yes` or `no`, indicating whether the person was inoculated or not. The variable `result` has outcomes `lived` or `died`,

		inoculated		Total
		yes	no	
result	lived	238	5136	5374
	died	6	844	850
	Total	244	5980	6224

Write out, in formal notation, the probability that a randomly selected person who was not inoculated died from smallpox, and find this probability.

$P(\text{died} \mid \text{not inoculated})$

A: died

B: not inoculated

$$P(A \mid B) = \frac{P(A \overset{\text{and}}{\wedge} B)}{P(B)} = \frac{844 / 6224}{5980 / 6224} = 0.14$$

Solution

$$P(\text{result} = \text{died} \mid \text{inoculated} = \text{no}) = \frac{P(\text{result} = \text{died and inoculated} = \text{no})}{P(\text{inoculated} = \text{no})} = \frac{(844/6224)}{(5980/6224)} = .141$$

Exercise

The people of Boston self-selected whether or not to be inoculated.

- a. Is this study observational or was this an experiment?
- b. Can we infer any causal connection using these data?
- c. What are some potential confounding variables that might influence whether someone lived or died and also affect whether that person was inoculated?

Recap

- Example & visual illustration of probability
- DeMorgan's Laws
- Conditional probability
- Independence and disjoint events

Next class

- Lecture 5: Medical diagnostics

Participation

- Name + How was class